

MeshEcho: Confidence-Aware Route Admission for LoRa Mesh IoT

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Abstract—LoRa mesh IoT networks trade source-confirmed reliability against scarce airtime. Flooding adds redundancy but creates collisions, whereas a cached source route can select a weak multi-hop path. We present MeshEcho, a confidence-ranked source-routing policy with bounded route-miss recovery, evaluated in a packet-level simulator. In 480 first-discovery unicasts over 20 held-out seeds, MeshEcho reaches 0.692 ACK completion at 93.3 s airtime, versus 0.554/91.4 s for ETX. The seed-paired ACK difference is +0.137 (95% CI [+0.090,+0.185]), at +1.9 s airtime. The primary policies do not always observe the same routes: only 148/480 MeshEcho-ETX candidate sets match. An isolated experiment with identical candidate sets on all 480 pairs retains a +0.163 ACK advantage [+0.121,+0.204], at +0.094 s per pair. On unconditioned random pairs, flooding instead achieves 0.736 ACK PDR/39.7 s against MeshEcho’s 0.430/103.2 s. Short route TTL under block fading also degrades MeshEcho to 0.341/71.1 s. These results support a conditional route-selection benefit in simulation, not universal dominance or hardware validity.

Index Terms—LoRa mesh, Internet of Things, route admission, packet delivery ratio, airtime

I. Introduction

LoRa is attractive for low-power IoT because it combines long link budget with inexpensive radios [1]. A star topology is insufficient for obstructed or infrastructure-poor deployments, motivating multi-hop extensions [2]–[4]. Multi-hop forwarding is expensive: a low data rate occupies the channel for a long time, while a flood multiplies collision opportunities. A cached source route reduces redundancy, but one weak hop can invalidate an otherwise useful path.

MeshEcho records lightweight link-quality evidence during discovery and ranks candidate paths by confidence. The default confidence threshold is zero, so an admitted route does not trigger confidence-based flooding; bounded TTL-2 recovery occurs only when discovery finds no candidate. The contribution is a route-selection policy evaluated against shortest-candidate, ETX, ETT, min-hop, and flooding baselines. The implementation is a simulation prototype, not a production mesh stack.

This paper makes four contributions:

- a reproducible packet-level LoRa mesh harness with time-on-air, propagation, shadowing, half-duplex collision/capture, route discovery, and energy accounting;
- a quality-gated, 480-unicast first-discovery matrix over 20 held-out seeds, with paired confidence intervals and candidate-set auditing;

- an isolated single-flow comparison with identical observed candidate sets, plus component and controlled route-conflict evidence; and
- random-pair, three-hop, and fading diagnostics that expose limits of the simulated reliability–airtime tradeoff.

II. Related Work

Prior ICC LoRa studies include analytical and simulation work on coverage, spreading-factor allocation, and frame error rates [5]–[9], as well as testbed validation [10], [11]. Multihop studies expose the flooding–airtime tradeoff [2], [4] and show that routing state and packet timing matter [12], [13]. ETX is a standard expected-transmission metric [14]; our fixed-SF ETT row is a sanity baseline because constant packet time-on-air makes it numerically coincide with ETX. We therefore ask a narrower question: does confidence admission change the reliability–airtime operating point under one shared packet model?

III. Protocol and Model

A. Network model

Nodes are uniformly placed in a square and share one half-duplex LoRa channel. Unless noted, packets have 32-byte payloads, 17 dBm transmit power, 125-kHz bandwidth, coding rate 4/5, capture threshold 6 dB, and SF7. Airtime follows the standard LoRa symbol calculation [15]. Log-distance path loss with log-normal shadowing produces SNR; reception is a logistic function of SNR margin, and overlapping packets are accepted only when the desired signal passes capture. The simulator records collisions, route-cache events, discovery success, and configured TX/RX energy.

B. Routing policies

Managed flooding rebroadcasts with duplicate suppression. Matched source-routing floods a request, collects candidates arriving in a 2-s window, and selects the shortest returned candidate. ETX and ETT use the same discovery and forwarding machinery but minimize

$$C_{\text{ETX}} = \sum_i p_i^{-1}, \quad C_{\text{ETT}} = \sum_i \text{ToA}_i p_i^{-1},$$

where p_i is the observed per-hop PRR. The min-hop baseline ignores link quality after the same discovery-window configuration. MeshEcho selects the candidate with the

highest confidence, using path length only as a tie-break. Its default low-confidence fallback threshold is zero; only discovery with no candidate invokes TTL-2 recovery. The ICC runs disable adaptive timeout retries. The historical Smart-CALM firmware/simulator namespace remains in the repository but is not an ICC protocol, ablation, or conclusion.

C. Confidence update and recovery

For each received route request, let m_i be the received SNR minus the SF-specific demodulation threshold on hop i , and let I_i indicate a reception marked as collided. The link score and MeshEcho’s accumulated candidate score are

$$\ell_i = \text{clip}_{.05}^{.98} \left(\frac{1}{1 + e^{-.7m_i}} - .18I_i \right), \quad (1)$$

$$c_1 = \ell_1, \quad c_i = \text{clip}_{.05}^{.98} (\min(c_{i-1}, \ell_i) - .025), \quad i \geq 2. \quad (2)$$

Here $\text{clip}_a^b(x) = \min(b, \max(a, x))$. Thus the score retains the weakest observed hop and charges 0.025 for each additional hop beyond the first. It is a heuristic admission score, not an independently calibrated end-to-end delivery probability. The destination retains every candidate arriving before the two-second window closes; the highest score wins, with shorter paths breaking equal-score ties. A route reply follows the reverse selected path, and data are sent only after the source receives that reply. Consequently, route-discovery success and ACK completion are distinct events, both included in the reported denominator.

For a cached route created at t_0 and expiring at t_e , forwarding uses

$$\hat{c}(t) = \text{clip}_0^1 \left[c_i - .1 \text{clip}_0^1 \left(\frac{t - t_0}{\max(1, t_e - t_0)} \right) \right]. \quad (3)$$

The default fallback threshold is zero, so this age reduction never launches a low-confidence duplicate for an admitted route. If discovery observes no candidate, the pending unicast instead invokes a TTL-2 recovery flood. ACK-triggered cancellation applies to delayed fallbacks, but the evaluated configuration disables timeout retry. This precise scope matters when interpreting the near-zero no-fallback ablation difference.

IV. Experimental Design

A. Primary matrix and metrics

The primary matrix has 50 nodes in an 8250 m square, 600-s runs, one unicast per selected directed pair (24 per seed), SF7, 4-dB shadowing, and held-out seeds 21–40. Pairs are selected from a shared analytical graph with edge $\text{PRR} \geq 0.90$ and graph distance 2–4 hops; every selected pair is two hops in these runs. The quality gate requires a complete pair pool, mean graph distance at least 2, and at least 10% of direct links below 0.99 PRR. Each pair appears once, so the estimand is first discovery rather than repeated-pair cache reuse. Policies share node placement,

TABLE I
Calibrated ICC parameters.

Nodes / area	50 / 8250 m × 8250 m
Duration / load	600 s / 24 one-shot unicasts per seed
Radio	32 B, 17 dBm, 125 kHz, CR 4/5, SF7
Path loss / shadowing	$n = 2.75$ / $\sigma = 4$ dB
Discovery / route	2 s / 600 s
TTL	
Pairs / graph	24 / $\text{PRR} \geq 0.90$, 2–4 hops
Seeds / quality gate	21–40 / mean hops ≥ 2 , below-0.99 PRR $\geq 10\%$

flow schedule, shadowing, reception-stream construction, discovery window, route TTL, and RREQ relay timing. Distinct forwarding decisions still alter collision histories and observed candidate sets. Reported paired intervals use 20 seed-level differences, not 480 independent flows.

B. Generalization strata

Separate 20-seed strata avoid pooling different estimands. Random pairs use 50 nodes in an 18-km square and mixed traffic at 4 flows/min. Deep multihop uses 100 nodes in a 21-km square and unicast at 2 flows/min; the selection rule permits graph distance 3–5, but all selected paths in these seeds are three hops. Temporal-fading cases reuse four connected pairs in the 8.25-km square with 6-dB block fading every 60 s, comparing 600-s and 30-s route TTLS. These cases are not pooled with Table II.

C. Comparison and uncertainty procedure

The primary comparison puts all 24 selected flows into one 600-s run for each policy and seed. The application schedule and topology are identical within a seed, and matched relay timing removes a known request-timing asymmetry. Nevertheless, path choice changes reply, data, and ACK transmissions; later discoveries can therefore see different interference and candidate sets. This matrix estimates complete-policy behavior under sequential operation. An audit pairs discoveries by seed, source, destination, and request identifier, then compares exact candidate-path sets rather than their counts alone.

To separate candidate ranking from this feedback, the isolation experiment runs each of the same 24 directed pairs alone in a fresh 12-s simulator for each of four policies: MeshEcho, ETX, min-hop, and matched source routing. All runs reuse their seed’s node geometry, radio settings, and reception stream, begin the unicast at 0.1 s, share matched request relay timing, and disable MeshEcho’s route-miss fallback. No previous flow can alter a route cache or the collision history. Candidate-set equality is verified for each pair before interpreting selection differences. The pair pool comes from an analytical connectivity graph, but no policy sees that graph when discovering a route.

For a baseline B , let d_s be MeshEcho’s metric minus B ’s metric in topology seed s . In the matrix, a PDR metric is the fraction of 24 observed unicasts; in isolation it is

TABLE II

One-shot first-discovery matrix (20 seeds, 480 unicasts). Values are seed means; $\pm$ denotes 95% CI half-width. ACK PDR requires a reverse-path ACK, while destination PDR does not.

Protocol	ACK PDR	Dest. PDR	Airtime (s)	Discovery	Multi-hop route	Mean hops
Managed flooding	.444 $\pm$.049	.992 $\pm$.010	27.8 $\pm$ 0.8	n/a	n/a	n/a
Matched source-route	.425 $\pm$.061	.494 $\pm$.058	90.5 $\pm$ 0.4	.633	.678	1.73
ETX metric	.554 $\pm$.051	.604 $\pm$.048	91.4 $\pm$ 0.4	.767	.717	1.76
Fixed-SF ETT metric	.554 $\pm$.051	.604 $\pm$.048	91.4 $\pm$ 0.4	.767	.717	1.76
Min-hop	.475 $\pm$.062	.537 $\pm$.063	90.8 $\pm$ 0.4	.696	.653	1.69
MeshEcho	.692 $\pm$.053	.708 $\pm$.057	93.3 $\pm$ 0.4	.829	.997	2.06

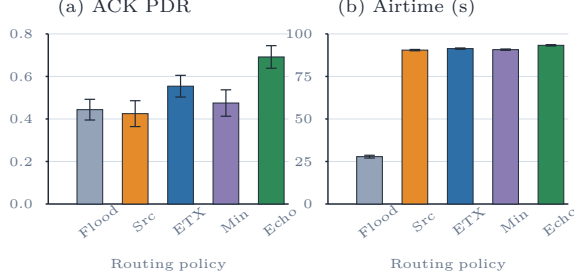


Fig. 1. One-shot matrix over $n = 20$ topology seeds. Bars are seed-level means and whiskers are two-sided 95% confidence intervals. The policy order is managed flooding, matched source-route (Src), ETX/ETT, min-hop (Min), and MeshEcho (Echo). ETX and fixed-SF ETT coincide under the shared packet-time model and are shown as one metric bar.

TABLE III

MeshEcho component attribution in the one-shot matrix. Deltas are MeshEcho minus the named variant; intervals are paired 95% CIs.

Variant	Δ ACK	95% CI	Δ airtime
No confidence	+ .212	[+.160, +.265]	+2.4 s
No fallback	+ .002	[-.002, +.006]	+0.0 s
No hop penalty	+ .010	[-.037, +.058]	-1.1 s
No age penalty	+ .000	[+.000, +.000]	+0.0 s

the mean of 24 paired binary outcomes within that seed. We report $\bar{d}$ and the two-sided interval $\bar{d} \pm t_{.975, 19} s_d / \sqrt{20}$, where s_d is the sample standard deviation of the 20 seed differences. Airtime is per 600-s run in the matrix and per isolated pair in the ranking test. Treating 480 flows as independent replicates would give unjustifiably narrow intervals because pairs in one seed share geometry and shadowing.

V. Results

A. Primary and mechanism evidence

MeshEcho exceeds ETX/ETT by +0.137 ACK PDR (95% paired CI [+0.090, +0.185]), with +1.9 s airtime ([+1.6, +2.3]). Against matched source routing it gains +0.267 ACK PDR ([+0.208, +0.325]) at +2.8 s airtime ([+2.4, +3.2]). Destination-PDR gains also remain positive against ETX (+0.104, [+0.051, +0.157]). Flooding is much cheaper in airtime (27.8 s), but its ACK PDR is 0.444 in this conditioned matrix; its destination PDR is 0.992, underscoring the reverse-ACK distinction.

TABLE IV

Isolated first discovery: means over 480 pairs, with identical candidate sets for all four policies. Airtime is per pair.

Policy	ACK PDR	Dest. PDR	Airtime (s)
Matched source-route	.433	.496	3.762
Min-hop	.458	.517	3.769
ETX	.525	.573	3.791
MeshEcho	.688	.706	3.884

TABLE V

ACK-PDR/airtime means in separate 20-seed strata.

Case	MeshEcho	ETX	Δ ACK	Δ airtime
Random pairs	.430/103.2	.293/103.0	+ .137	+0.2 s
Three-hop, 100 nodes	.724/154.9	.551/151.6	+ .172	+3.3 s
Fading, TTL 600 s	.581/21.8	.483/20.8	+ .098	+1.1 s
Fading, TTL 30 s	.341/71.1	.235/54.8	+ .106	+16.3 s

The no-confidence row is an end-to-end bundle comparison: candidate sets, discovery success, and route length can all change. In the main matrix, MeshEcho and ETX observe identical candidate sets on only 148/480 paired flows, despite matched RREQ timing. Thus Table II alone cannot attribute the gain to route ranking. A separate isolated first-discovery experiment restarts the simulator for each pair, disables route-miss fallback, and obtains identical candidate sets for all 480 pairs; 447 pairs have at least two candidates. MeshEcho gains +0.163 ACK PDR over ETX ([+0.121, +0.204]), at +0.094 s per pair ([+0.078, +0.110]). When both select the same path (169 pairs), ACK, destination, and airtime outcomes match. This conditional result supports a route-choice mechanism under the static model. A controlled seven-node route-conflict audit also finds a +0.192 ACK gain (± 0.100) for the longer selected branch at +14.3 s airtime; it is not evidence of universal superiority.

B. Generalization and temporal fading

Random pairs retain a +0.137 ACK advantage over ETX ([+0.097, +0.178]), but managed flooding instead reaches 0.736 ACK PDR at 39.7 s, against MeshEcho's 0.430 at 103.2 s. The deeper case admits a 3–5-hop selection rule, yet all selected graph paths are three hops in the observed seeds; MeshEcho gains ACK PDR over ETX at +3.3 s airtime. Under temporal fading, shortening TTL from 600 to 30 s increases MeshEcho's mean route repairs

TABLE VI
Secondary temporal-fading diagnostic (20 seeds).

TTL	ACK PDR	Dest. PDR	Airtime (s)	Repairs
600 s	.581	.619	21.8	0.0
30 s	.341	.356	71.1	14.5

from 0 to 14.5 per run, reduces ACK PDR from 0.581 to 0.341, and increases airtime from 21.8 to 71.1 s. Flooding reaches 0.561/44.7 s in the short-TTL workload. These are operating-point boundaries, not evidence that confidence aging solves fading.

C. Fairness and metric boundaries

The matched source-route baseline uses the same discovery window and RREQ relay timing as MeshEcho, but differing transmissions still change candidate exposure. The isolated experiment is the stronger fixed-candidate check. We report ACK and destination PDR separately: an ACK confirms source-side completion, whereas destination delivery can remain high when the reverse path fails. Airtime includes discovery, data, ACK, and recovery traffic.

The fading table is not a replacement for the one-shot primary matrix: it reuses four pairs and measures route-cache aging. Pair conditioning, static shadowing, and the reverse-path ACK model remain separate validity limits.

VI. Discussion and Limitations

The primary matrix shows higher ACK completion for MeshEcho than ETX at a small airtime cost, but not a clean route-ranking effect: only 148/480 MeshEcho-ETX candidate sets match. The isolated experiment, with all candidate sets matched, supports a conditional route-choice interpretation. The budgeted variant falls to 0.573 ACK PDR, below default MeshEcho’s 0.692; it is an unsuccessful tradeoff here, not an improvement. The no-fallback ablation differs by only +0.002 ACK PDR in the matched matrix, consistent with fallback having little effect in this workload. Random-pair flooding and short-TTL fading reverse the attractive operating point, so the result is not topology-independent dominance.

All results are packet-level simulation. The model includes path loss, shadowing, collision timing, capture, route discovery, and energy accounting, but omits waveform interference, oscillator offset, modem implementation timing, duty-cycle enforcement, variable route headers, and antenna effects. The confidence estimate and forwarding channel share the same realization in static runs; temporal block fading is a deterministic stress model, not hardware validation. A hardware or hardware-in-the-loop study remains necessary for radio-specific external validity.

A. Threats to interpretation

The quality gate is a measurement contract, not a guarantee of deployment realism. Analytical-graph pair

selection removes disconnected samples but conditions the primary estimate on usable static links. Random pairing removes that conditioning and markedly changes the ranking; the deep stratum tests a different scale, with actual selected graph paths all three hops. These strata must not be averaged into one ranking.

The paired design shares topology, traffic, and reception-stream construction, but it is not event-by-event common-random-number coupling; different transmissions cause different collision histories. Confidence intervals use independent seed-level differences, not individual flows. The no-confidence ablation changes discovery as well as route selection, and the seven-node route-conflict audit is deliberately surgical. Neither replaces the isolated same-candidate comparison.

B. Deployment boundary

The simulated link score uses an SNR margin from successful request receptions. It does not explicitly estimate packet loss from censored requests, nor does it calibrate confidence against measured end-to-end delivery. Static shadowing lets discovery and forwarding sample the same link realization. A moving node, changing antenna orientation, or external interferer could invalidate the ordering before the 600-s route TTL expires. The short-TTL fading case probes one such failure mode, but its 60-s block process is not a measured channel trace. The lower ACK rate and higher airtime in that case show that merely rediscovering more often is not a complete mobility solution.

A radio study should keep the selected pair schedule and two-second discovery window fixed while logging every request candidate, chosen path, per-hop RSSI/SNR, data arrival, reverse ACK, retry, and transmit airtime. Repeating the isolated same-candidate test on a small SX126x or SX127x testbed would test whether the score ordering persists under real oscillator offset, antenna variation, modem timing, and interference. A second phase should impose an enforceable duty-cycle budget and vary route TTL, then report both ACK and destination PDR. Without those measurements, the present numbers should guide simulator comparisons rather than deployment claims.

VII. Reproducibility

Version 2.1.23 includes source, tests, seed-level CSVs, summary CSVs, candidate-path records, and reports. The matrix, isolation, and candidate audit utilities live in the released tools/ directory; their exact arguments and output prefixes are recorded alongside the raw results. Generalization, fading, and route-conflict runners are versioned separately. The 20 held-out seeds are 21–40; raw CSVs preserve per-seed denominators and selected paths. No human participants or personal data were involved, and the authors report no external funding or conflicts of interest.

TABLE VII
Version 2.1.23 evidence strata and estimands.

Artifact	Main perturbation	Estimand
One-shot	PRR-qualified, 2-hop pairs	System comparison
Isolated	Identical candidate sets	Route ranking
Random pairs	Unconditioned source/destination	Generalization
Deep multihop	100 nodes, observed 3 hops	Scale/hop depth
Fading	6-dB blocks, TTL 600/30 s	Staleness boundary

A. Evidence manifest

Raw observations are separate from interpretation. The primary CSV contains one row per policy and seed and records 24 scheduled and observed distinct unicasts in each row. Discovery records preserve observed candidate paths and selected routes, allowing the 148/480 live candidate-set match to be recomputed. The isolated CSV records each policy’s result for every pair; the report checks all 480 candidate sets and same-path outcomes. Generalization reports keep the random-pair, deep, and fading estimands separate. Historical 2.1.22 sensitivity artifacts remain labeled as historical and are not pooled with this holdout.

B. Audit checklist

Before interpretation, the release checks protocol coverage, paired traffic traces, candidate exposure, route TTL, and independent reception streams. Connected-pool runners reject incomplete pair sets and degenerate link regimes. Temporal fading is a deterministic function of link, seed, and time block. Reports include selection rules, route-hop diagnostics, discovery success, and paired intervals, so high aggregate PDR cannot hide a single-hop or discovery-limited experiment.

The raw files retain negative cases: flooding dominates MeshEcho on random pairs, a short TTL is costly under fading, and the budgeted variant loses ACK PDR. Fixed-SF ETT and ETX coincide because the packet airtime multiplier is constant. CSVs, not rounded table entries, are authoritative for recomputing intervals.

VIII. Conclusion

Across 480 one-shot unicasts, MeshEcho attains 0.692 ACK PDR/93.3 s versus ETX’s 0.554/91.4 s. The isolated same-candidate experiment supports a route-ranking benefit, while the live matrix also changes candidate exposure. Random-pair flooding outperforms MeshEcho, and short-TTL fading substantially degrades it. Version 2.1.23 therefore supports a bounded simulation result, not universal dominance or physical validation.

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